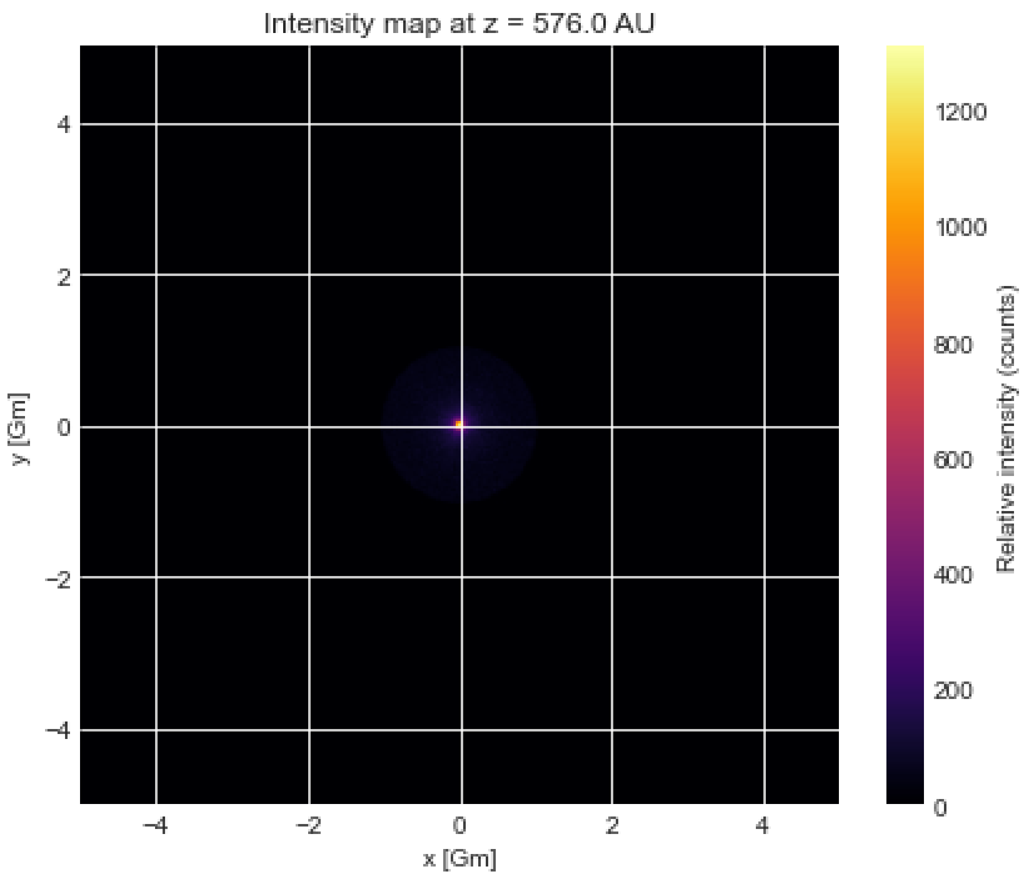
The code I put on uses Newton-Raphson to come up with an optimised focal length (i.e. how far away the receiver should ideally be) for the Sun, for the highest intensity. So this is specifically for where we should put a probe to receive signals from far away I think! I have plotted the history of how the NR code developed to show how it reached the result.

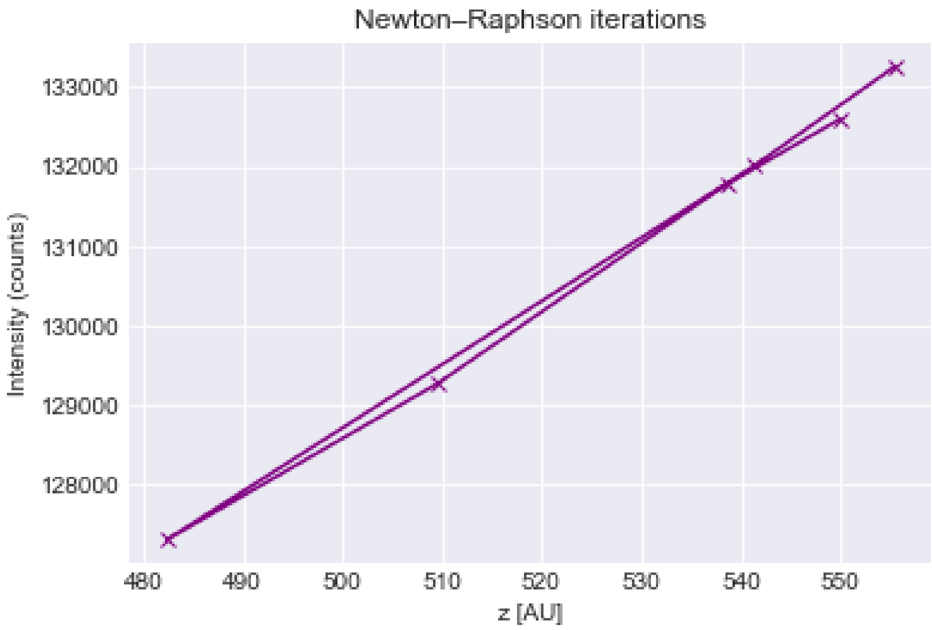
Monte-Carlo is also used to simulate a high number of random rays coming in to like get the intensity profile that we would receive, and have also coded something close to Emily's to get an intensity profile picture.

Runs can come up with quite different results (bc monte carlo can give such varied results unless u run it enough for my laptop to explode… that or the code doesn't work BUT I THINK IT DOES) but from online we know it should be around 550 AU and it does get close to that – we only need An attempt that gets us close to that for the poster anyways!

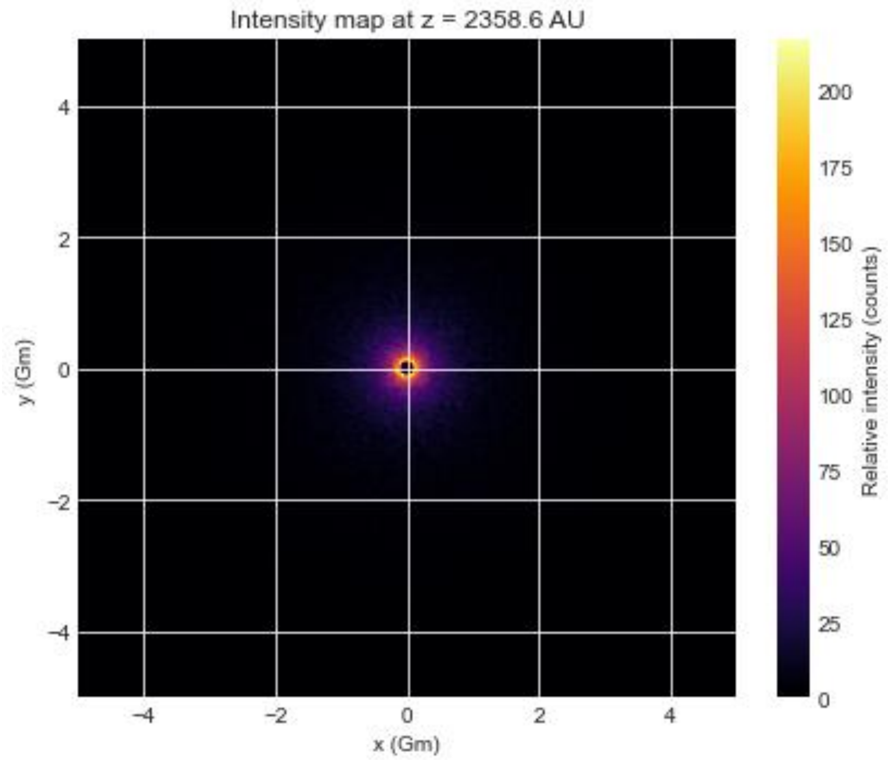
I think it works? Required a lot of inspiration from other code online but at least to me the logic checks out … 'Counts' refers to the amount of simulated beams hitting the receiver – not anything of value irl but I guess proportional to power or amplitude of the signal(?)

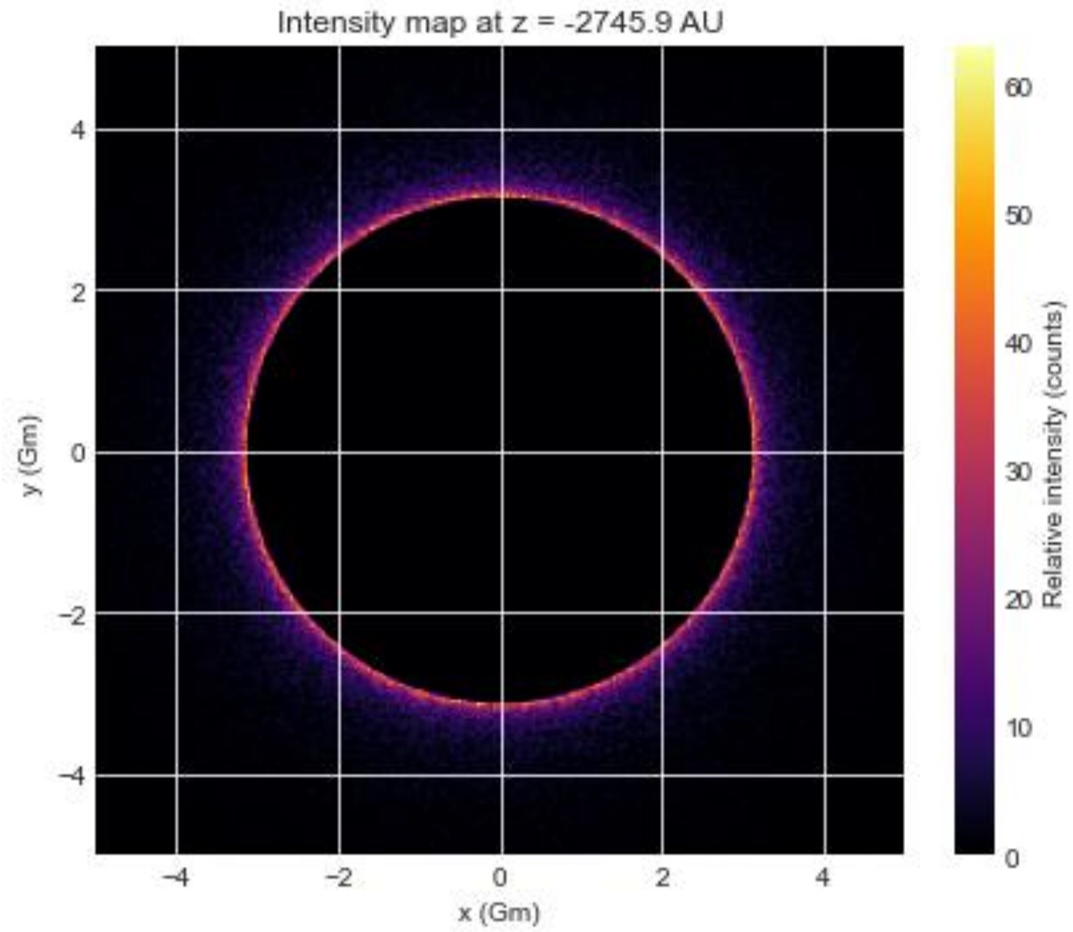
This is an example where it got 576AU (pretty close) :





You get some cool shapes for focal distances where the code has bugged out (too high, negative etc) also:

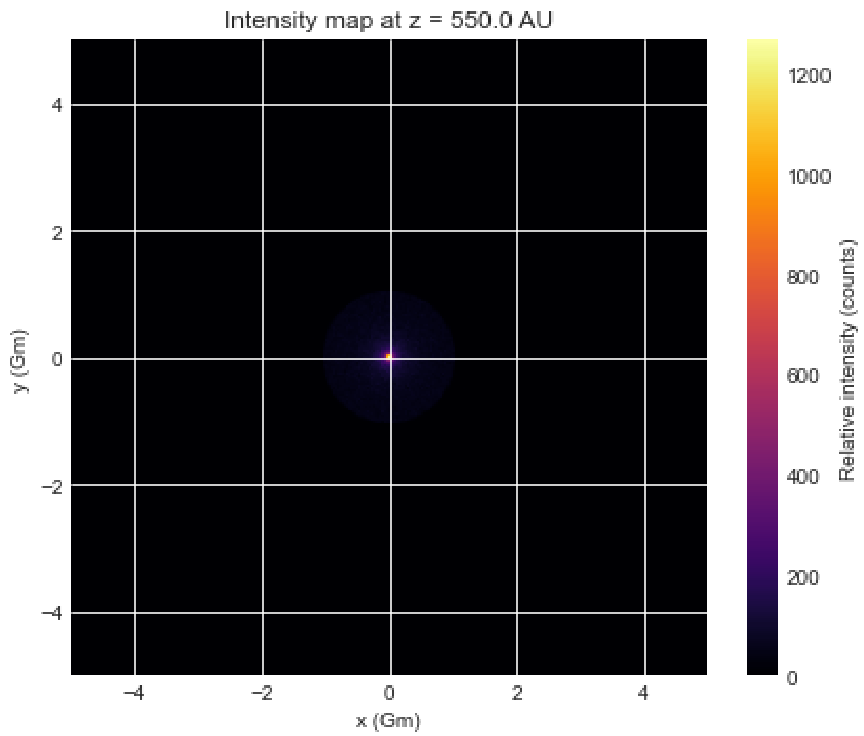




Negative one maybe interesting as negative focal length means image is behind lens hence the black circle would be the blocking from the sun / the object or something idk?

ALSO. If unconvinced with the Newton-Raphson stuff, if we just used 550 AU (justifying with either a reference or using other arguments) and ran the plotting code, that still uses Monte-Carlo!

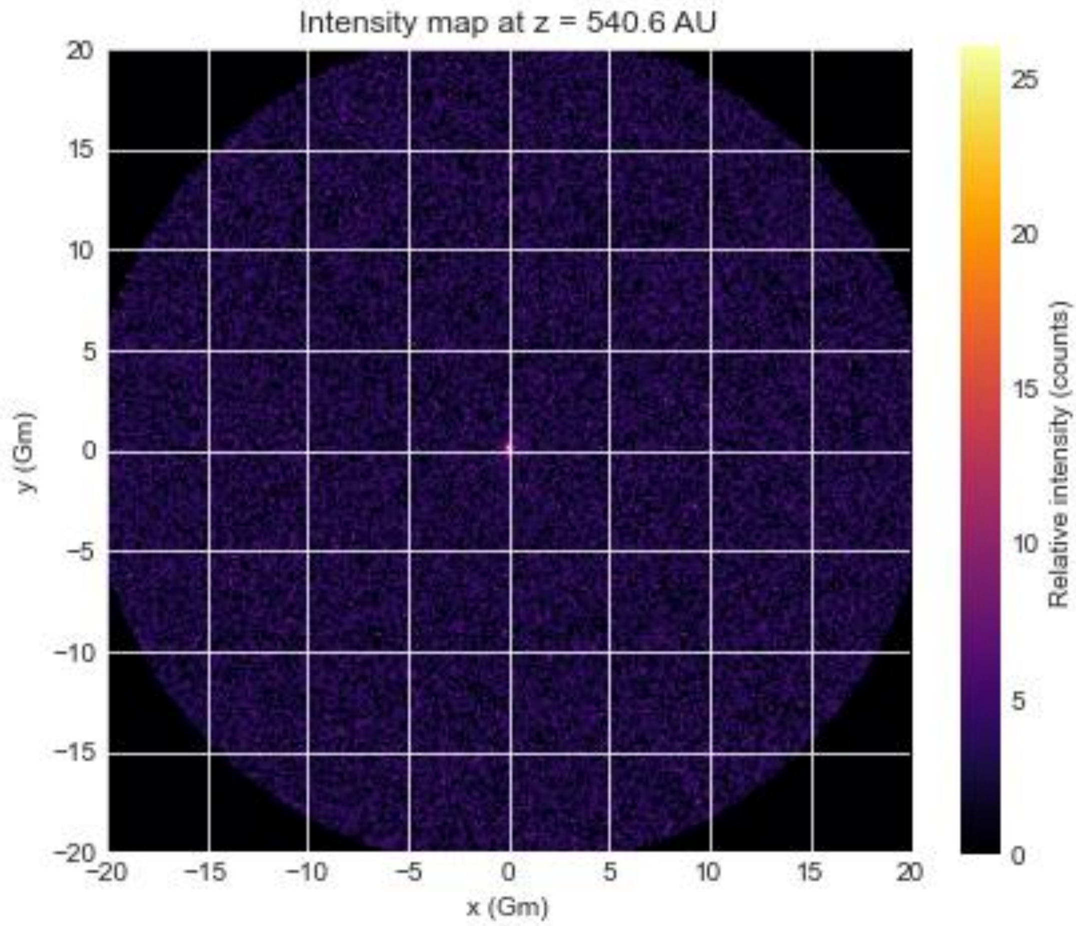
This is 550:



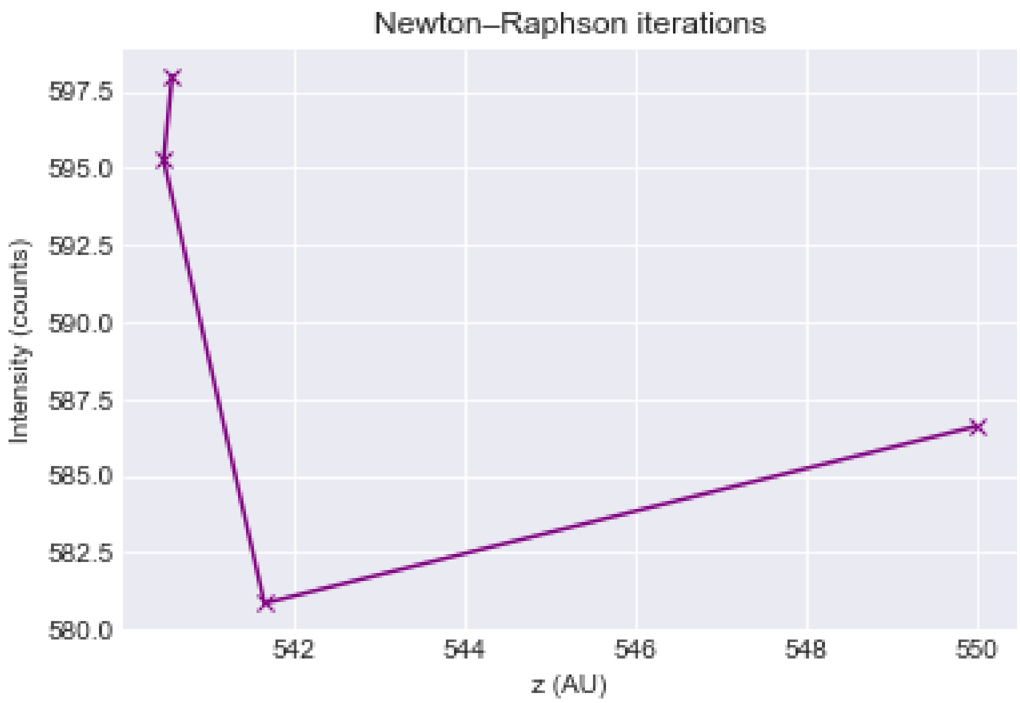
UPDATE

Have worked out a way to make it more consistent by increasing the impact parameter maximum (ends up w more noise but still w focussed spot in middle), and then calculating an average of the intensities over 20 attempts, and increasing the size of the axes in the intensity map.

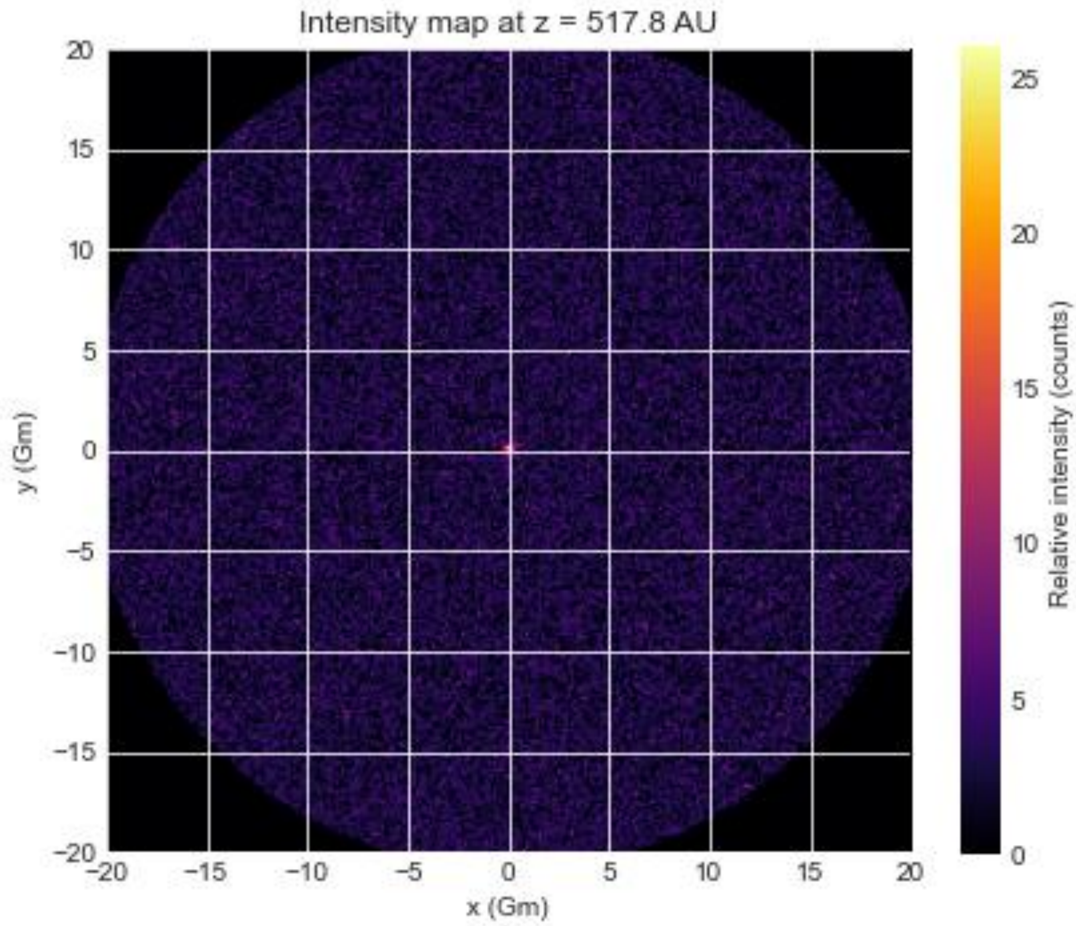
5 diff runs of the code w impact param as 30 * Rsun are all consistent all close to whats expected:

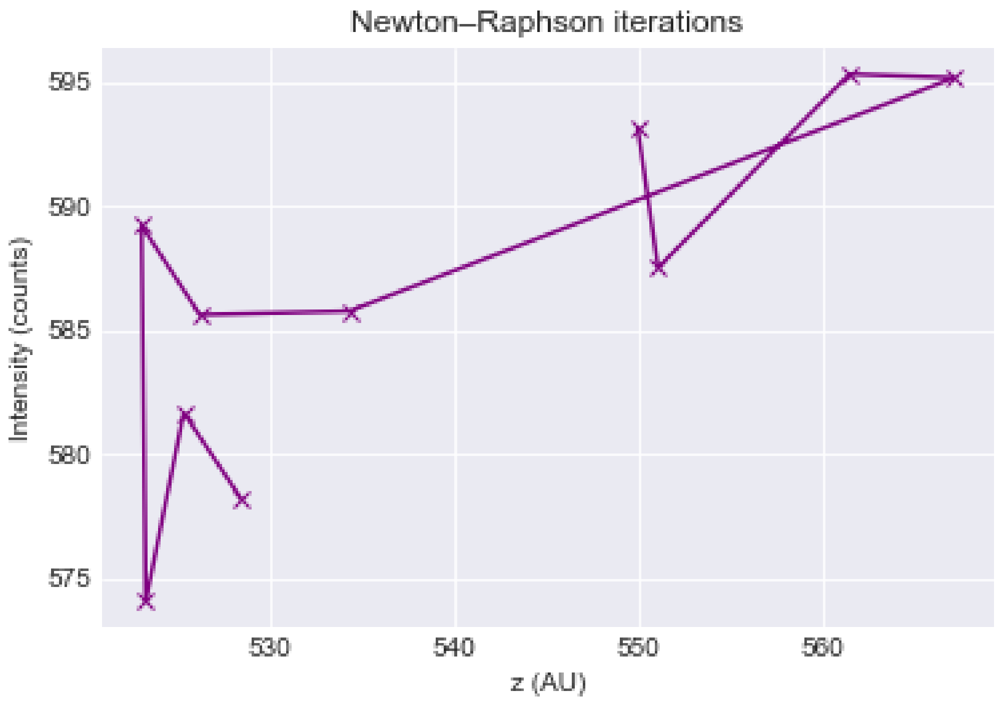


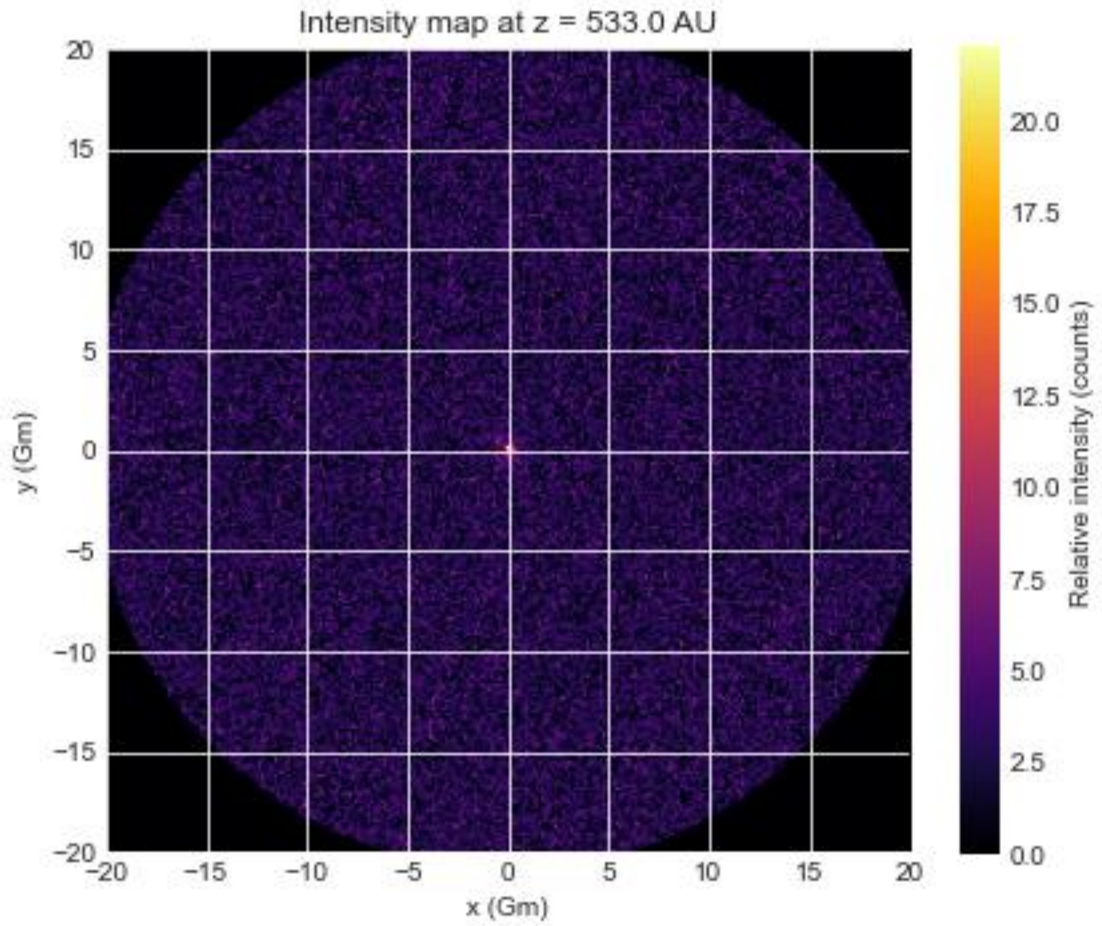
Intensity map at z = 540.6 AU
y (Gm)
x (Gm)
Relative intensity (counts)
20
15
10
5
0
−5
−10
−15
−20
25



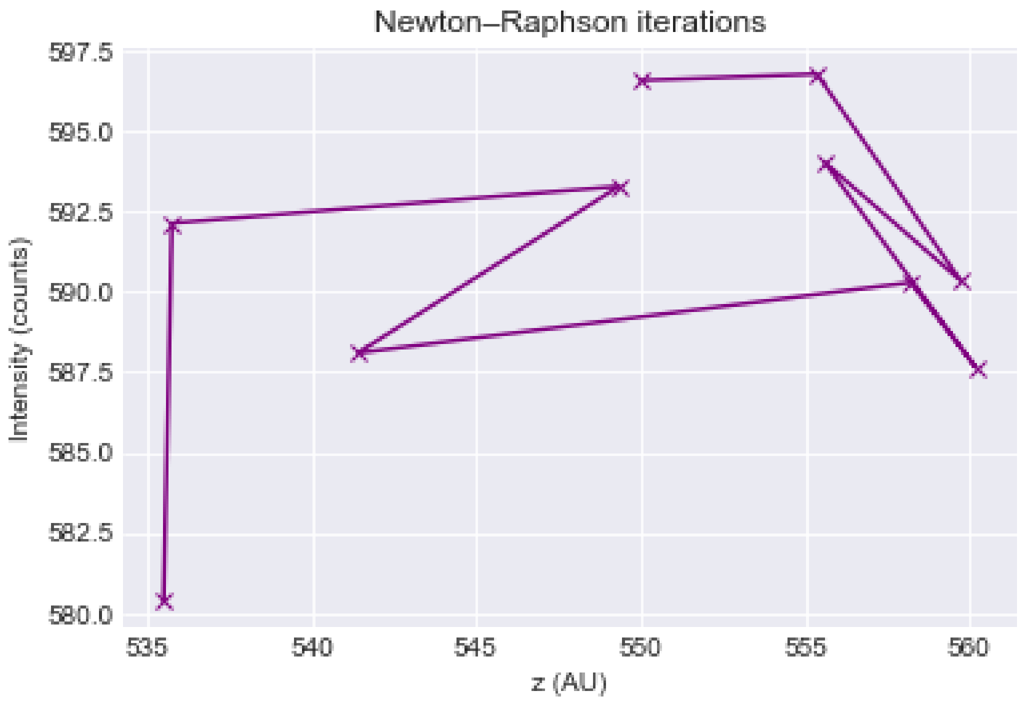
Newton–Raphson iterations
Intensity (counts)
z (AU)
597.5
595.0
592.5
590.0
587.5
585.0
582.5
580.0
542
544
546
548
550


Intensity map at z = 517.8 AU
y (Gm)
x (Gm)
Relative intensity (counts)
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
25
20
15
10
5
0

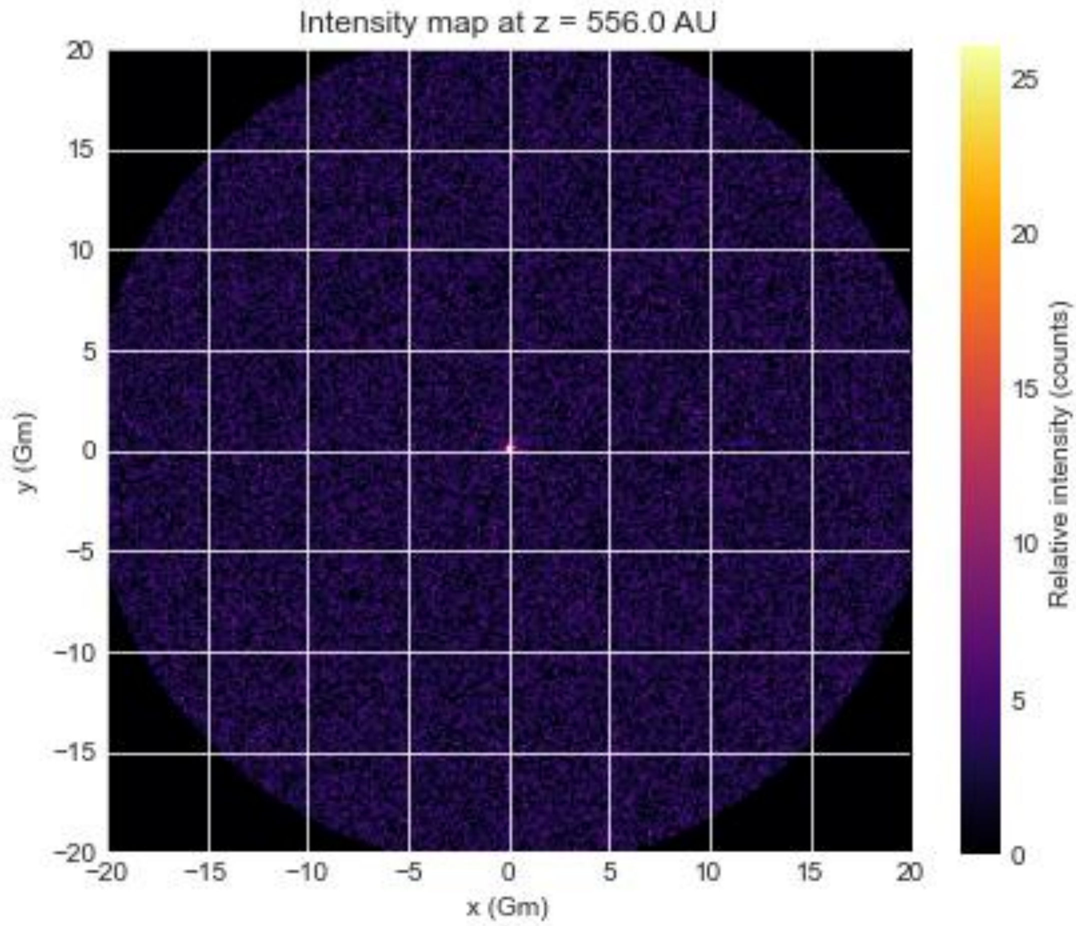

Newton–Raphson iterations
Intensity (counts)
z (AU)
595
590
585
580
575
530
540
550
560



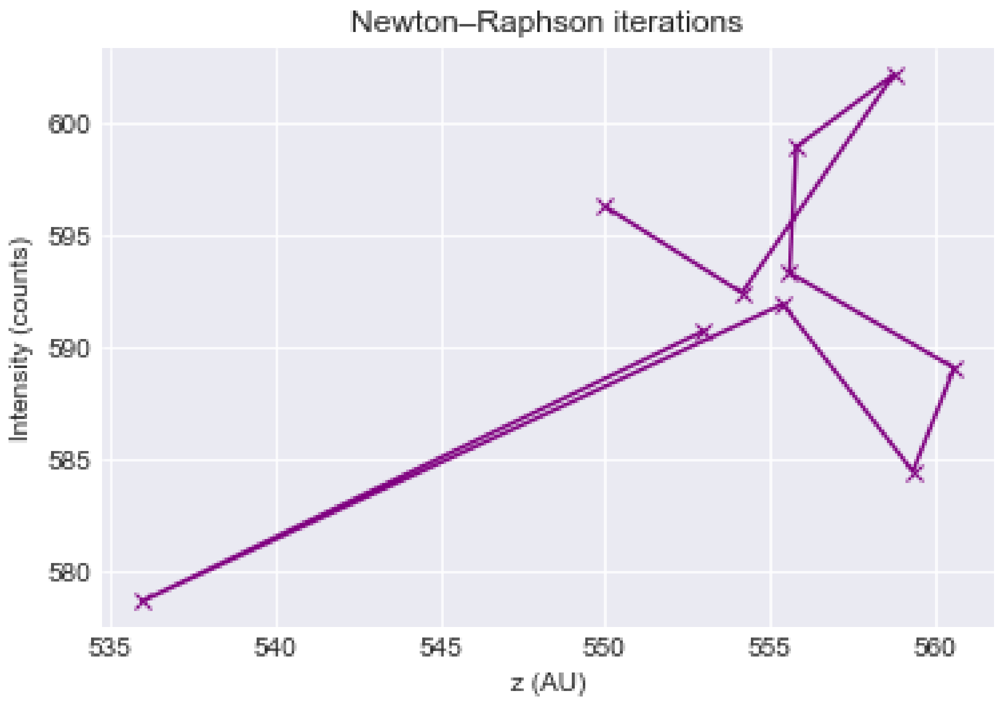
Intensity map at z = 533.0 AU
y (Gm)
x (Gm)
Relative intensity (counts)
20
15
10
5
0
−5
−10
−15
−20
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0



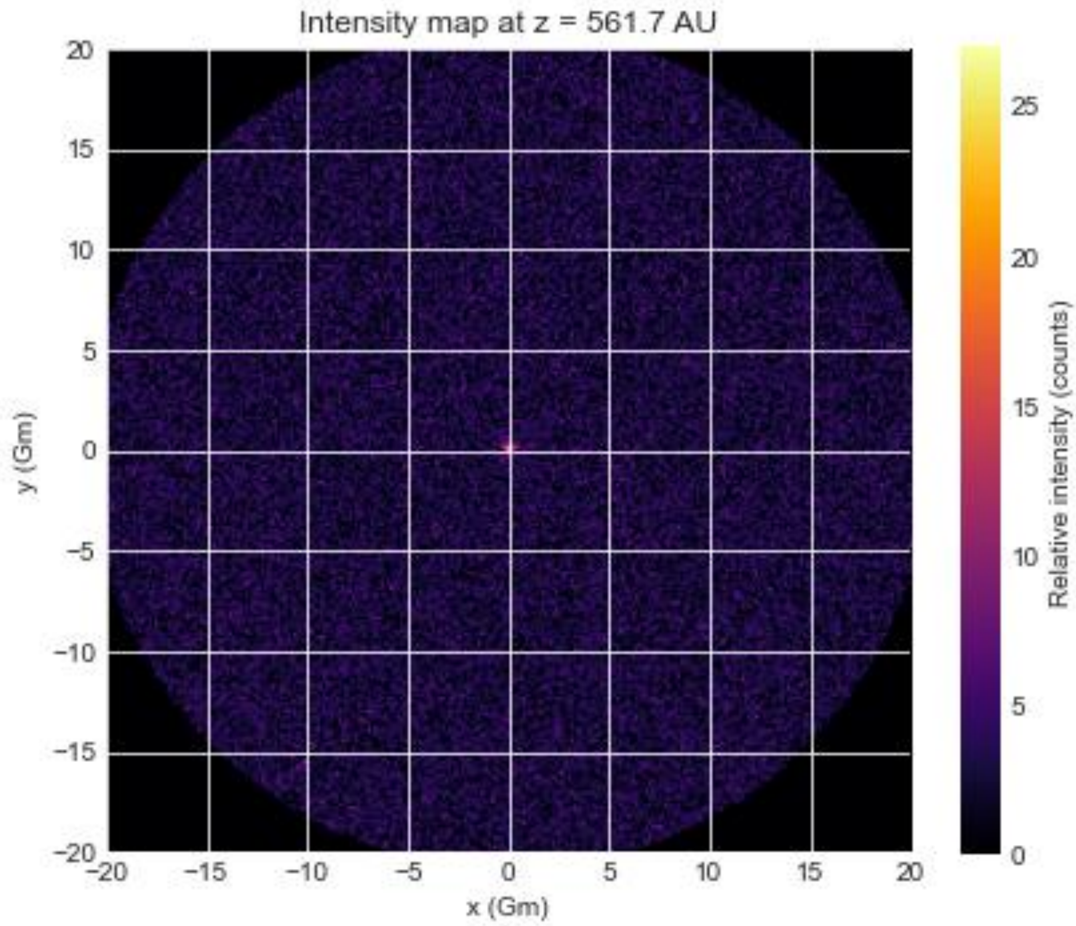
Newton–Raphson iterations
Intensity (counts)
z (AU)
597.5
595.0
592.5
590.0
587.5
585.0
582.5
580.0
535
540
545
550
555
560

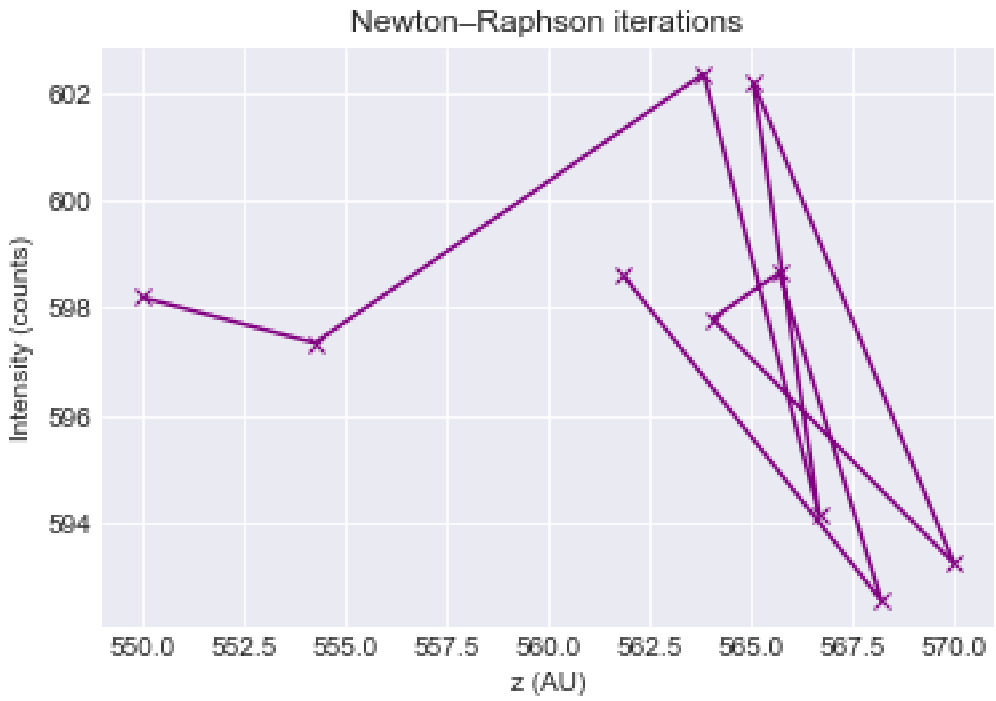


Intensity map at z = 556.0 AU
y (Gm)
x (Gm)
Relative intensity (counts)



Newton–Raphson iterations
Intensity (counts)
z (AU)


Intensity map at z = 561.7 AU
y (Gm)
x (Gm)
Relative intensity (counts)
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
25
20
15
10
5
0


Newton–Raphson iterations
Intensity (counts)
z (AU)
602
600
598
596
594
550.0
552.5
555.0
557.5
560.0
562.5
565.0
567.5
570.0

BASICALLY

- minimum distance that the receiver (z minimum) can be is 550AU because (from NASA powerpoint)

The gravitational deflection angle of light passing a massive body is

- $\theta = (4GM/c^2)(1/r)$ (1)

where r is the distance by which the rays being focused miss the center of the sun. The distance F to the gravitational focus is:

- $F = (4GM/c^2)^{-1} r^2$ (2)

The condition required to use of the gravitational lens as a telescope is that the light from the distant target reaching the focus must not intersect the surface of the sun, and hence r must be greater than the radius of the sun (about 700,000 km).

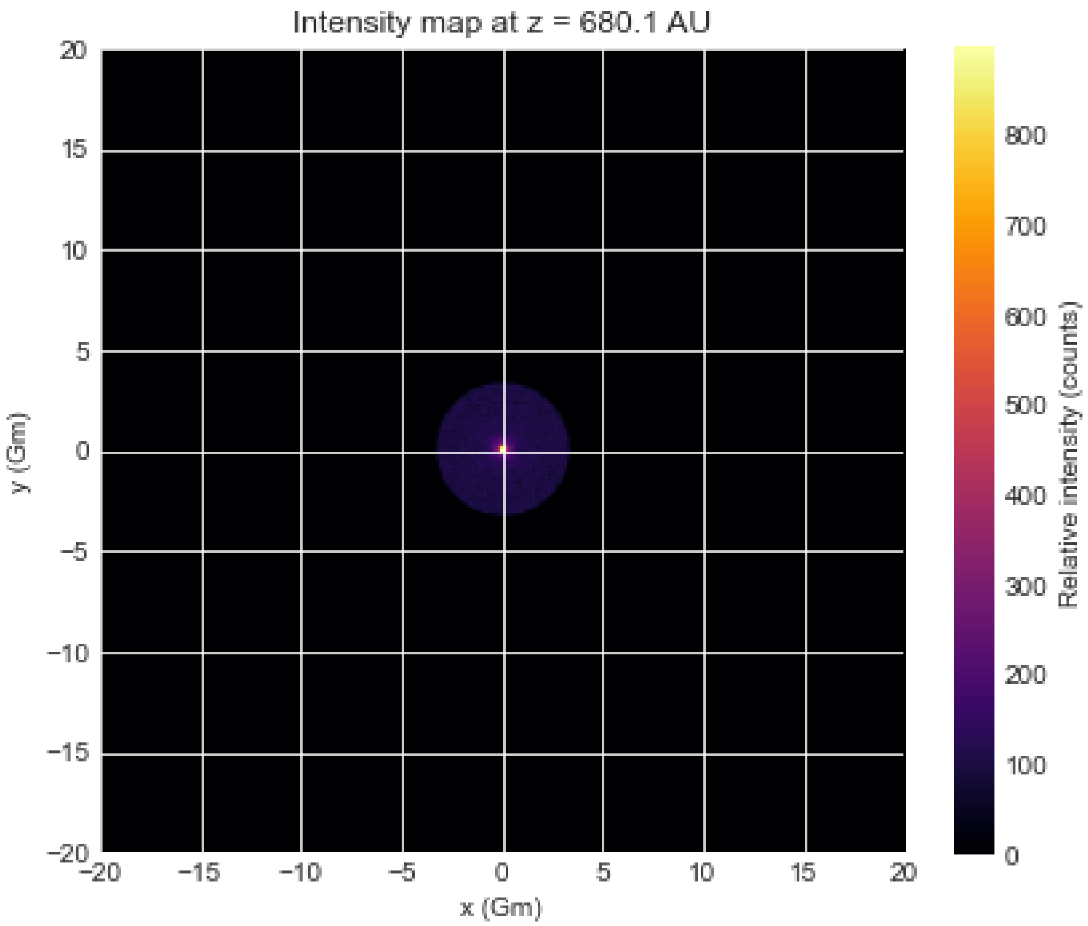
Inserting $r= R_{sun}$ defines the minimum gravitational focus.

For the sun, this minimum focal distance (at which the focussed rays just skim the surface of the sun) is ~550 AU

--any distance beyond this will **also** be on the focal line

- Hence this is either for us having sent a receiver to 550AU OR an alien at 550AU OR 2 alien civilisations, one of which at 550AU (where it would be a focal line not point).
- Distances >550AU possible, but further away = longer = loss of power (by inverse square), loss of quality etc
- This code is finding the optimal distance, by finding peak intensity ... should be near 550
- Source assumed to be very far away to allow for parallel rays incoming (i.e. basically infinity)
- Tolerance in the code is 0.1AU so this is the error I think?
- Assumes sender receiver and sun on same axis
- Earth to sun is 1AU... not as far as we would like for the parallel approx. but if narrow beam its alright... better to assume everything further away

- Lowering max impact parameter again so beams all closer to sun to feel its effect. Trying 5:



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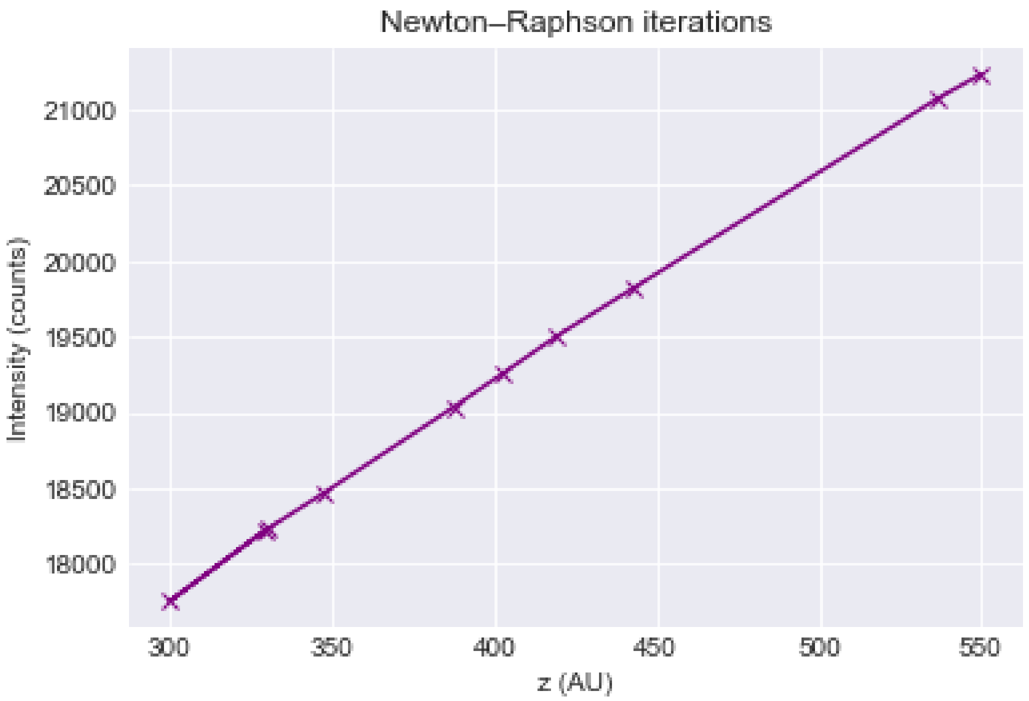
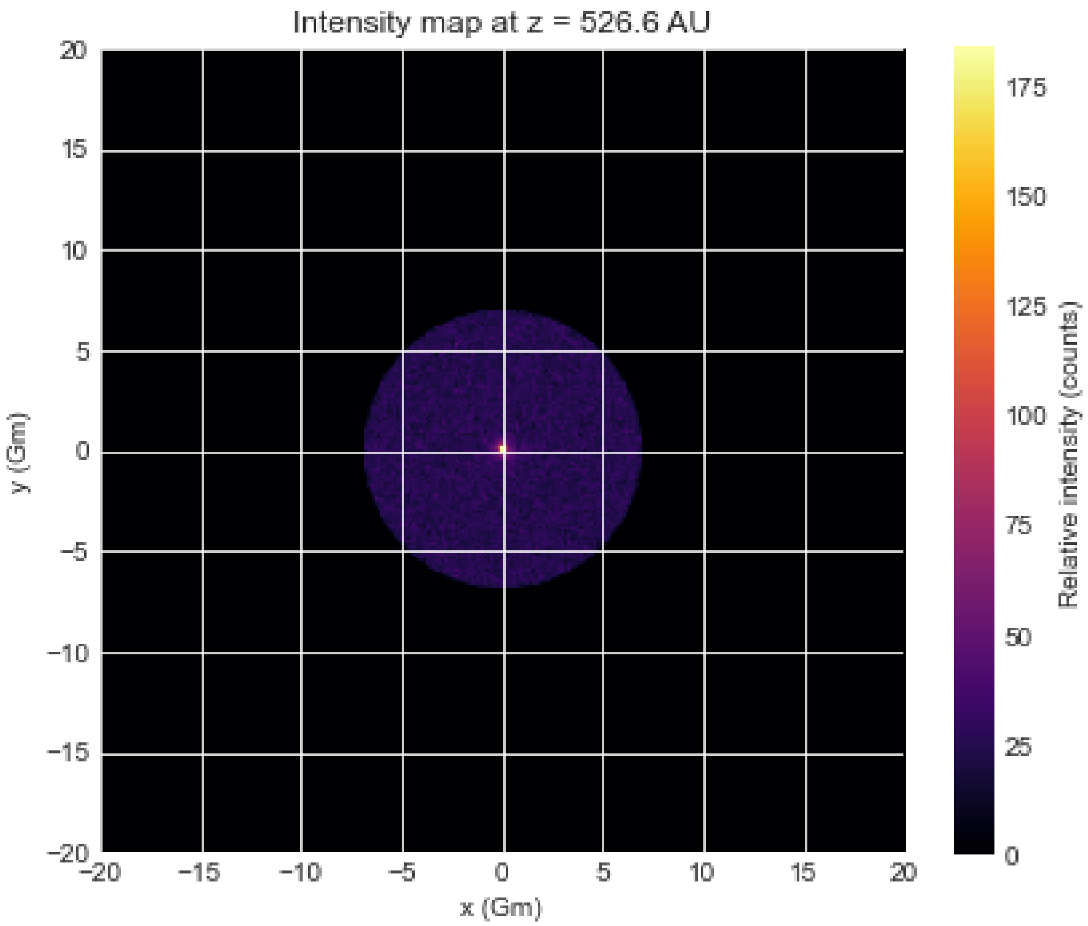
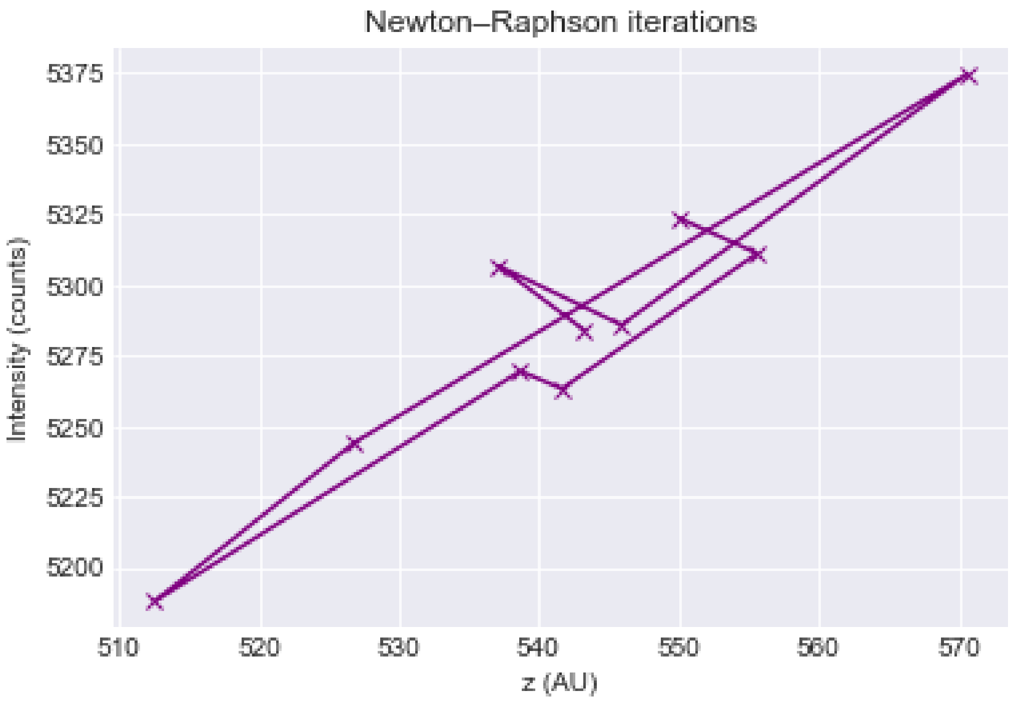


-
- Trying 10 Rsun

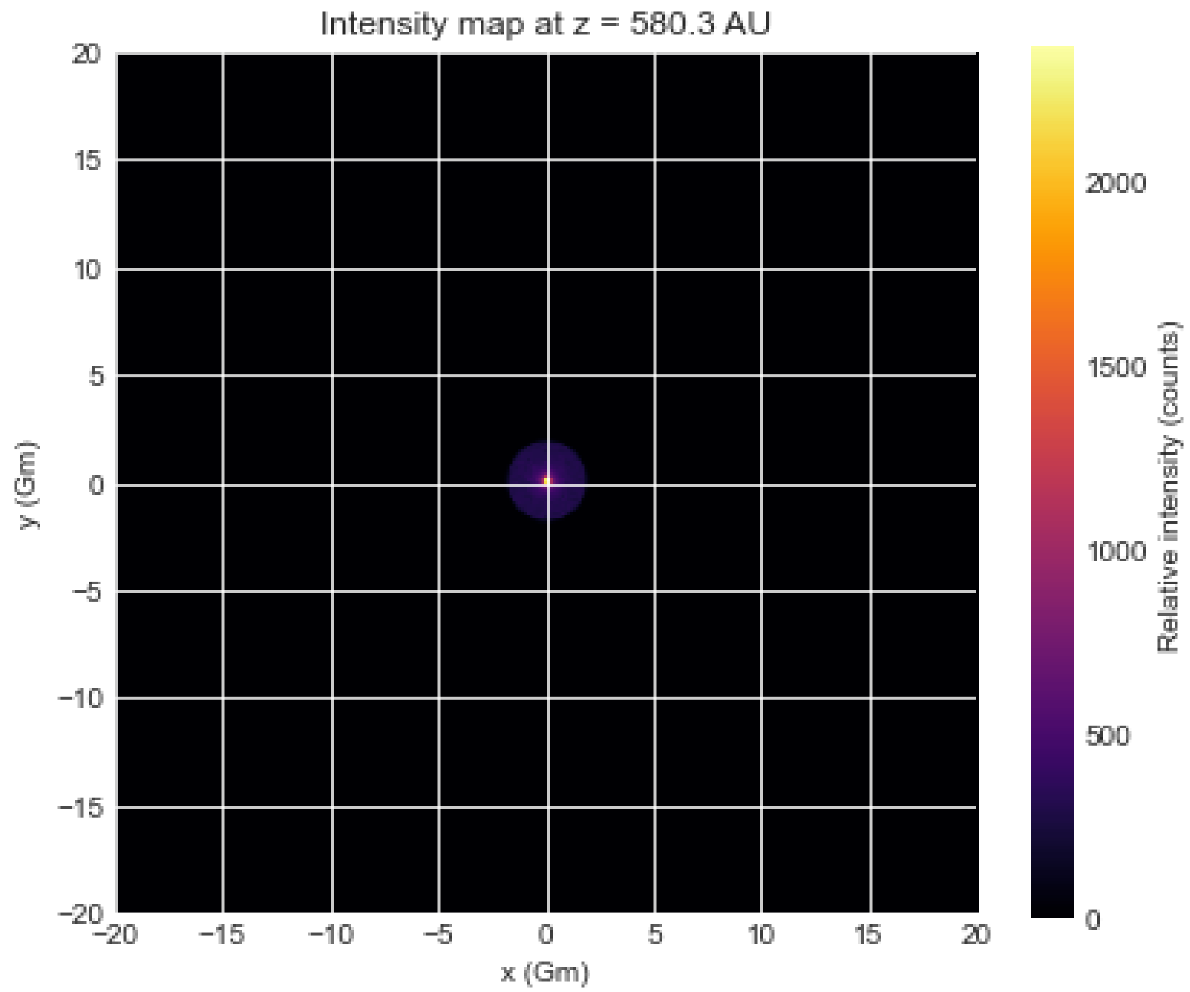

Intensity map at z = 526.6 AU
y (Gm)
x (Gm)
Relative intensity (counts)
20
15
10
5
0
−5
−10
−15
−20
175
150
125
100
75
50
25
0

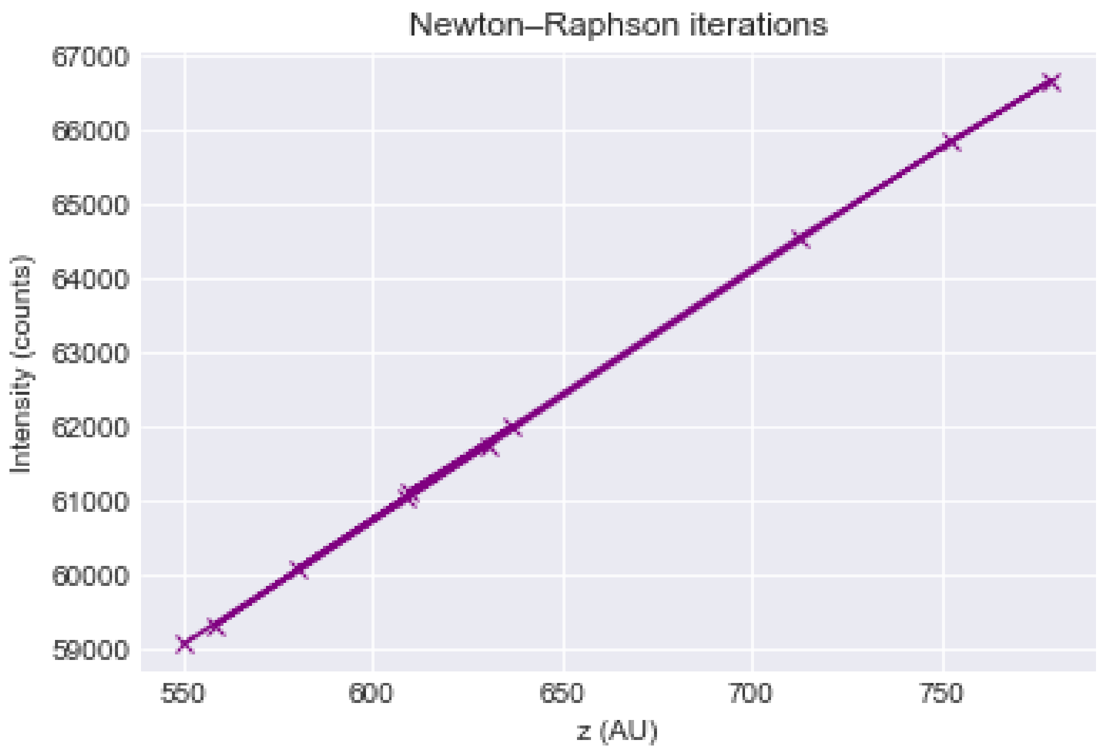
-


Newton–Raphson iterations
Intensity (counts)
z (AU)
5375
5350
5325
5300
5275
5250
5225
5200
510
520
530
540
550
560
570

-

Trying 3 Rsun * this ones pretty good maybe.**




Newton–Raphson iterations
Intensity (counts)
67000
66000
65000
64000
63000
62000
61000
60000
59000
550
600
650
700
750
z (AU)